Derbyshire Shared Care Pathology Guidelines

Hypercalcaemia

Purpose of Guideline
The management of patients with newly diagnosed hypercalcaemia

Definition
Adjusted (corrected) calcium > 2.60 mmol/L

Importance of Hypercalcaemia

Hypercalcaemia can be a medical emergency (see below). Even when calcium is not raised to acutely dangerous levels it is important to define the cause because hypercalcaemia can be a presenting feature of serious disease including malignancy.

When is hypercalcaemia considered a medical emergency?

- Calcium 2.60 – 3.00 mmol/L Not usually a medical emergency
- Calcium 3.00 – 3.50 mmol/L Possible medical emergency
- Calcium >3.50 mmol/L Usually a medical emergency

These limits are for guidance only; the severity of the patient's symptoms, and other factors, e.g. renal function, will also determine whether the patient requires emergency admission.

The laboratory usually telephones calcium results > 3.4 mmol/L to GP practices when open or to Derbyshire Health United when the GP practice is closed.

When is hypercalcaemia suspected?
Mild hypercalcaemia is relatively common, and many patients may be asymptomatic. The severity and number of symptoms will be dependent on the degree of hypercalcaemia. Patients who have developed hypercalcaemia acutely are also much more likely to have symptoms.

Symptoms include;

- Polyuria and polydipsia
- Nausea and vomiting
- Constipation
- Weakness and tiredness
- Depression and psychiatric disturbances
- Drowsiness and confusion
- Bone Pain
- Renal Stones

Polyuria and polydipsia are the most specific symptoms but presentation is often non-specific. Drowsiness is a late sign in terms of hypercalcaemia.

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Reviewed by: Dr P Blackwell, GP at Hannage Brook Medical Centre; Dr D Hughes & Dr A Ugur, Consultant Endocrinologists, RDH; Dr R Robinson, Consultant Endocrinologist, CRH; Mrs H Seddon, Clinical Scientist UHDB.
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What are the major causes of hypercalcaemia?

- Primary Hyperparathyroidism
- Linked to Renal Disease
  - Tertiary Hyperparathyroidism
  - Treatment with Vitamin D analogues
- Malignancy
  - Primary Solid Tumour
  - Secondary in Bone
  - Haematological Malignancy

>90% of cases

Less common causes of hypercalcaemia

- Hyperthyroidism
- Sarcoid
- Vitamin D Toxicity
- Other Causes, including drugs (Lithium, Thiazide)

Uncontrolled when printed

Corrected Calcium
2.51 – 2.60 mmol/L
WITH renal stones / osteoporosis or other significant and persistent clinical features of primary hyperparathyroidism

Corrected Calcium
2.61 – 2.80 mmol/L

Repeat calcium

1. Corrected calcium >2.5 mmol/L on 2 separate occasions AND renal stones / osteoporosis?

OR

2. Corrected calcium >2.6 mmol/L on 2 separate occasions?

No → Suggest re-check calcium dependent on clinical details / history

Yes → **Request paired calcium (gold top tube) and PTH (purple top tube)**

Corrected Calcium
> 2.80 mmol/L

Dependent upon clinical details / history and availability of EDTA sample, the laboratory measures PTH.
If EDTA tube not available suggest urgent repeat paired calcium (gold top tube) and PTH (purple top tube)

PTH >3.2 pmol/L

PTH-Dependent Hypercalcaemia

Primary Hyperparathyroidism is by far the commonest cause of PTH-dependent hypercalcaemia.

Differential diagnosis:
- Primary hyperparathyroidism (sporadic)
- Familial
  - Multiple Endocrine Neoplasia (MEN I & IIa)
  - Familial Hypocalciuric Hypercalcaemia (FHH)
- Tertiary hyperparathyroidism (advanced kidney disease)
- Medications (Lithium)

Whatever the PTH and calcium value:-
- Check & correct excessive dietary calcium
- Check prolonged inactivity
- Check if on Thiazide, ***and stop***

If calcium ≥3.00 mmol/L: Urgent referral to Endocrinology is recommended

If calcium <3.00 mmol/L: See Shared Care Primary Hyperparathyroidism guideline for ‘When to refer’

PTH 1.6 - 3.2 pmol/L

Equivocal

PTH neither clearly suppressed nor in range expected with primary hyperparathyroidism.
Suggest repeat paired calcium (gold top tube) and PTH (purple top tube)

PTH <1.6 pmol/L

Non PTH-Dependent Hypercalcaemia

Differential diagnosis:
- Malignancy
- Hyperthyroidism
- Vitamin D toxicity
- Sarcoid

See p4 of guideline for more detail

PTH-Dependent Hypercalcaemia

- **If calcium ≥3.00 mmol/L:** Urgent referral to Endocrinology is recommended
- **If calcium <3.00 mmol/L:** See Shared Care Primary Hyperparathyroidism guideline for 'When to refer'
- For patients with acute or chronic kidney disease a renal referral is recommended (acute renal impairment or new CKD5 needs urgent referral).
- For patients with lithium-dependent hypercalcaemia it is not recommended to stop the lithium treatment immediately, and each patient should be considered on a case by case basis. Recommend obtaining Endocrine advice by choosing 'Advice and Guidance' from NHS e-Referrals or see contacts at the end of this guideline.

Non PTH-Dependent Hypercalcaemia

Malignancy

Three main types:

- Solid Tumours, PTHrP related, e.g. Lung CA
- Haematological Malignancy, e.g. Myeloma
- Skeletal Metastases, e.g. Secondary to Breast CA

For patients with known cancer suggest urgent referral to patient's hospital team. When the underlying malignancy is not known, referral will obviously depend upon underlying symptoms, history and other findings that are available at the time.

Hyperthyroidism

For hypercalcaemia associated with hyperthyroidism, suggest urgent referral to Endocrinology.

Vitamin D toxicity

Hypercalcaemia caused by "over the counter" Vitamin D preparations is not common, and patients need to have a 25-Hydroxy Vitamin D level measured.
Most patients with hypercalcaemia caused by Vitamin D analogues (e.g. calcitriol) should be referred to the hospital team who first prescribed the medication.

Other causes

Unless the patient already has a definitive diagnosis (e.g. Sarcoid, or lymphoma) it is worthwhile referring patients with hypercalcaemia who do not fall into any of the above categories to Endocrinology.

A final point

Primary Hyperparathyroidism is a relatively common disorder, therefore some patients may have more than one aetiology for the hypercalcaemia and/or have an underlying malignancy present.

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131
Endocrinology Advice (urgent)	07879 115507 (9am – 5pm, Mon – Fri)
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
Endocrinology Advice (urgent)	01246 513104
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Burton

Duty Biochemist	01283 566333 ext 4233
Endocrinology advice	Via endocrine secretaries 01283 593259

References

1. Hyperparathyroidism (primary): diagnosis, assessment and initial management. NICE NG132 (May 2019)

Author: Dr Nigel Lawson, August 2011		
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Dr R Rea, Dr P Blackwell, Mrs H Seddon	Oct 2013	31st Oct 2015
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